

1202SCG Calculus 1
1020ENG Engineering Mathematics 2

Week 3 Lecture

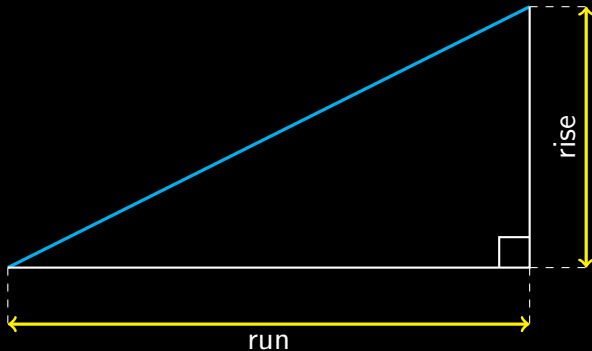
David Harman

Rates Are Everywhere

Many everyday quantities describe how one quantity changes relative to another.

- ▶ **Speed:** kilometres/hour
- ▶ **Pay:** \$/year or \$/hour
- ▶ **Medication:** tablets/day
- ▶ **Heart rate:** beats/minute
- ▶ **Water flow:** litres/minute
- ▶ **Data transfer:** megabytes/second (megabits/second)
- ▶ **COVID infection growth:** new infections/day

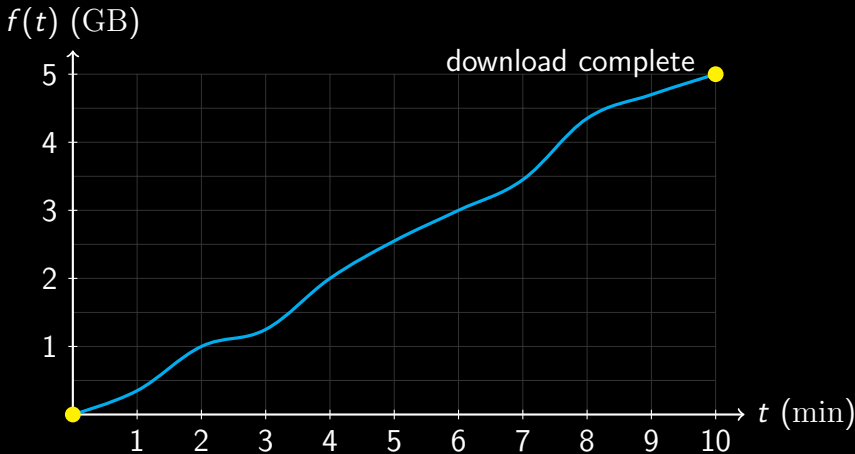
Revision of Gradient



$$\text{gradient} = \frac{\text{rise}}{\text{run}}$$

Downloading a Large File

Let $f(t)$ be the amount of a 5 GB file downloaded after t minutes.



Average Rates of Change

An average rate of change is measured between two points.

$$\text{Average rate of change} = \frac{f(b) - f(a)}{b - a},$$

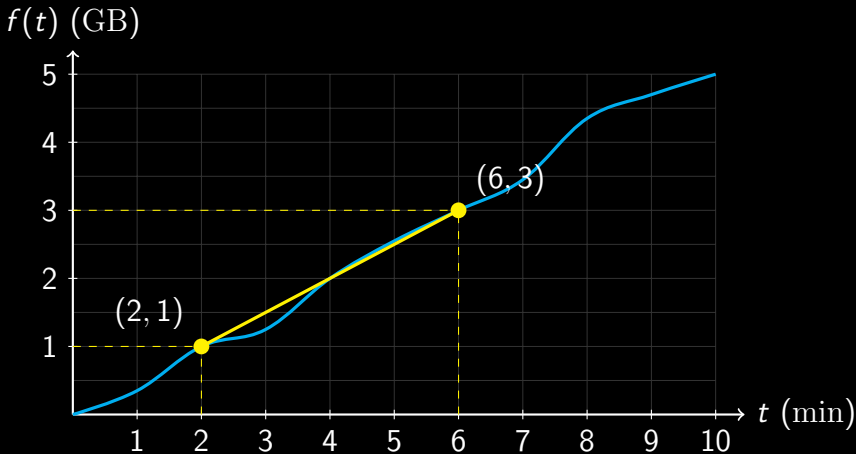
$$m = \frac{y_2 - y_1}{x_2 - x_1},$$

$$\text{Average download speed} = \frac{f(t_2) - f(t_1)}{t_2 - t_1}.$$

The average rate of change is the slope of the line joining the two points.

Average Download Speed

Calculate the average download speed between $t = 2$ and $t = 6$ minutes.



Average Download Speed: Solution

Between $t = 2$ and $t = 6$,

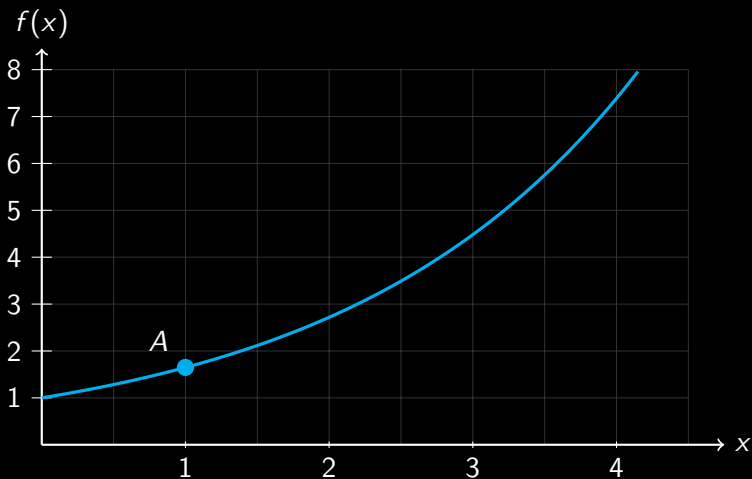
$$f(2) = 1 \text{ GB}, \quad f(6) = 3 \text{ GB}.$$

$$\begin{aligned}\text{Average download speed} &= \frac{f(6) - f(2)}{6 - 2} \\ &= \frac{3 - 1}{4} \\ &= 0.5 \text{ GB/min.}\end{aligned}$$

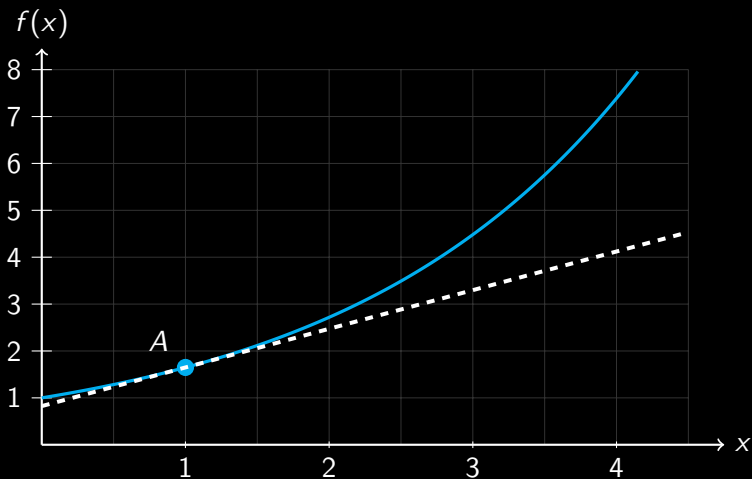
$$0.5 \text{ GB/min} = 500 \text{ MB/min} = 8.33 \text{ MB/s} = 66.7 \text{ Mb/s}.$$

Uppercase B denotes bytes; lowercase b denotes bits.

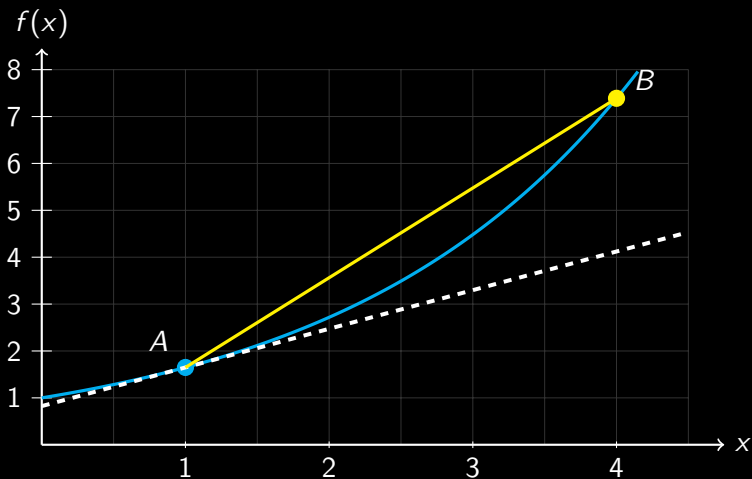
Rates of Change



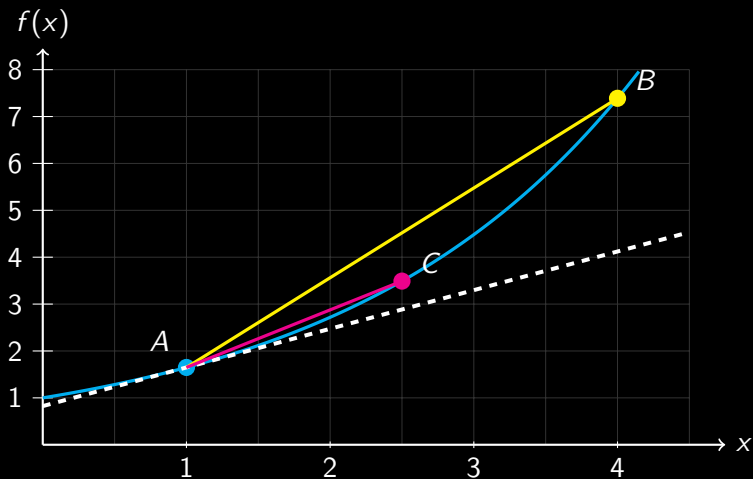
Rates of Change



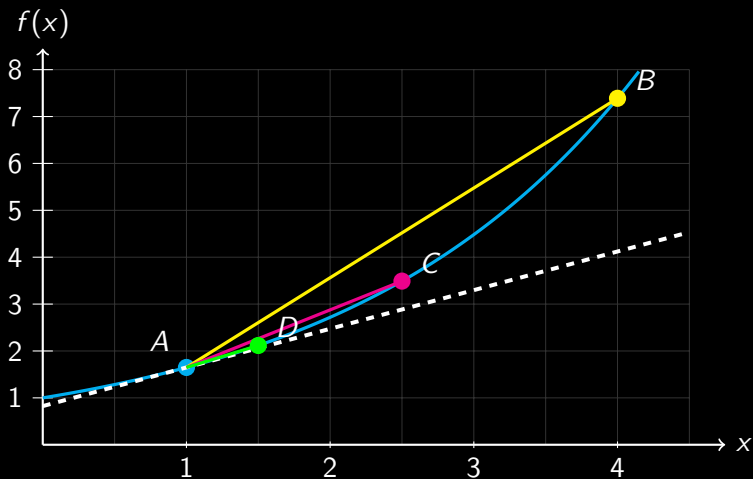
Rates of Change



Rates of Change

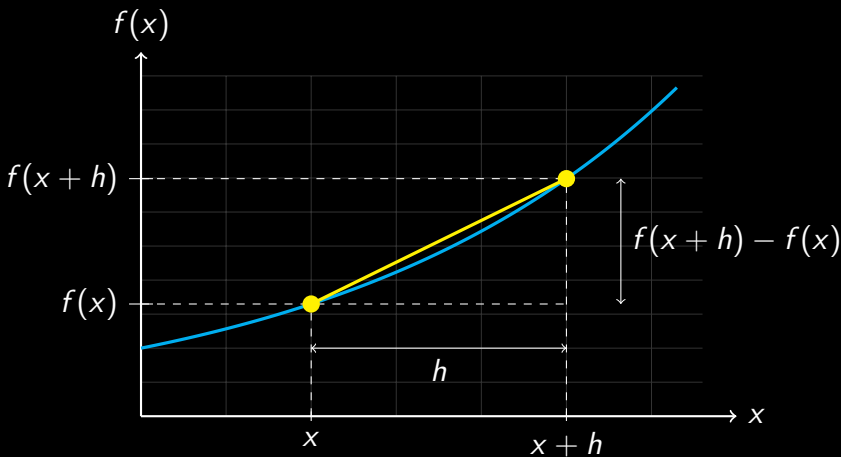


Rates of Change



Formal Definition of a Derivative

$$\frac{dy}{dx} = \lim_{h \rightarrow 0} \frac{f(x+h) - f(x)}{h}.$$



Derivative Notation

The first derivative may be written in several equivalent ways:

$$y', \quad f'(x), \quad \frac{dy}{dx}, \quad \frac{d}{dx}[f(x)].$$

Higher-Order Derivatives

Second derivative: $y'', \quad f''(x), \quad \frac{d^2y}{dx^2},$

Third derivative: $y''', \quad f'''(x), \quad \frac{d^3y}{dx^3}.$

Words That Mean Derivative

In this context, the following descriptions refer to the same idea:

- ▶ derivative;
- ▶ slope;
- ▶ gradient;
- ▶ instantaneous rate of change.

Each describes how quickly a quantity is changing at a particular point.

First Principles: $y = f(x) = x^2$

$$\begin{aligned}\frac{dy}{dx} &= \lim_{h \rightarrow 0} \frac{f(x+h) - f(x)}{h} \\&= \lim_{h \rightarrow 0} \frac{(x+h)^2 - x^2}{h} \\&= \lim_{h \rightarrow 0} \frac{x^2 + 2xh + h^2 - x^2}{h} \\&= \lim_{h \rightarrow 0} \frac{h(2x + h)}{h} \\&= \lim_{h \rightarrow 0} (2x + h) \\&= 2x.\end{aligned}$$

First Principles: $y = f(x) = 1/x$

$$\frac{dy}{dx} = \lim_{h \rightarrow 0} \frac{f(x+h) - f(x)}{h}$$

$$= \lim_{h \rightarrow 0} \frac{\frac{1}{x+h} - \frac{1}{x}}{h}$$

$$= \lim_{h \rightarrow 0} \frac{\frac{x - (x+h)}{x(x+h)}}{h}$$

$$= \lim_{h \rightarrow 0} \frac{-h}{hx(x+h)}$$

$$= \lim_{h \rightarrow 0} \left(-\frac{1}{x(x+h)} \right)$$

$$= -\frac{1}{x^2}.$$

Table of Derivatives

In the table below, n and c are constants.

Function y	Derivative $\frac{dy}{dx}$
c	0
x^n	nx^{n-1}
$\sin(x)$	$\cos(x)$
$\cos(x)$	$-\sin(x)$
e^x	e^x
$\ln(x)$	$\frac{1}{x}$

Using the Power Rule

Example 1

$$\begin{aligned}y &= \frac{1}{x^3} \\&= x^{-3}\end{aligned}$$

$$\begin{aligned}\frac{dy}{dx} &= -3x^{-4} \\&= -\frac{3}{x^4}.\end{aligned}$$

Example 2

$$\begin{aligned}y &= \sqrt{x} \\&= x^{\frac{1}{2}}\end{aligned}$$

$$\begin{aligned}\frac{dy}{dx} &= \frac{1}{2}x^{-\frac{1}{2}} \\&= \frac{1}{2x^{\frac{1}{2}}} \\&= \frac{1}{2\sqrt{x}}.\end{aligned}$$

Expanded Table of Derivatives

In the table, $a > 0$ and $a \neq 1$.

Function y	Derivative $\frac{dy}{dx}$
$\tan(x)$	$\frac{1}{\cos^2(x)} = \sec^2(x)$
a^x	$a^x \ln(a)$
$\log_a(x)$	$\frac{1}{x \ln(a)}$
$\sinh(x)$	$\cosh(x)$
$\cosh(x)$	$\sinh(x)$
$\tanh(x)$	$\frac{1}{\cosh^2(x)} = \operatorname{sech}^2(x)$

Rules for Combining Derivatives

Let c be a constant and suppose that $f(x)$ and $g(x)$ are differentiable.

Sum rule:
$$\frac{d}{dx}[f(x) + g(x)] = \frac{d}{dx}f(x) + \frac{d}{dx}g(x),$$

Difference rule:
$$\frac{d}{dx}[f(x) - g(x)] = \frac{d}{dx}f(x) - \frac{d}{dx}g(x),$$

Constant multiple rule:
$$\frac{d}{dx}[cf(x)] = c \frac{d}{dx}f(x).$$

Derivative Practice

Find $\frac{dy}{dx}$ for each function.

1. $y = 5x^7 - 3x^4 + 2x - 8$

2. $y = \frac{3}{x^4} - 2\sqrt{x}$

3. $y = 6 \sin(x) - 4 \cos(x)$

4. $y = 2 \tan(x) + 5x^3$

5. $y = 4e^x - 3(2^x)$

6. $y = 5 \ln(x) + 2 \log_3(x)$

7. $y = 3 \sinh(x) - 2 \cosh(x)$

8. $y = 4 \tanh(x) - 5x^2$

Derivative Practice: Answers

$$1. \frac{dy}{dx} = 35x^6 - 12x^3 + 2$$

$$2. \frac{dy}{dx} = -\frac{12}{x^5} - \frac{1}{\sqrt{x}}$$

$$3. \frac{dy}{dx} = 6 \cos(x) + 4 \sin(x)$$

$$4. \frac{dy}{dx} = 2 \sec^2(x) + 15x^2$$

$$5. \frac{dy}{dx} = 4e^x - 3(2^x) \ln(2)$$

$$6. \frac{dy}{dx} = \frac{5}{x} + \frac{2}{x \ln(3)}$$

$$7. \frac{dy}{dx} = 3 \cosh(x) - 2 \sinh(x)$$

$$8. \frac{dy}{dx} = 4 \operatorname{sech}^2(x) - 10x$$

Position, Velocity, and Acceleration

A delivery robot moves along a straight corridor. Its position is

$$s(t) = t^3 - 3t^2 + 2t + 5,$$

where s is measured in metres and t in seconds.

Determine:

1. the position of the robot at $t = 2$ and $t = 3$;
2. its average velocity between $t = 2$ and $t = 3$;
3. its instantaneous velocity at $t = 2$;
4. its instantaneous acceleration at $t = 2$.

Position and Average Velocity

$$\begin{aligned}s(2) &= (2)^3 - 3 \times (2^2) + 2(2) + 5 \\ &= 5 \text{ m}\end{aligned}$$

$$\begin{aligned}s(3) &= (3)^3 - 3 \times (3^2) + 2(3) + 5 \\ &= 11 \text{ m}\end{aligned}$$

$$\begin{aligned}\text{Average velocity} &= \frac{s(3) - s(2)}{3 - 2} \\ &= \frac{11 - 5}{1} \\ &= 6 \text{ m/s.}\end{aligned}$$

Instantaneous Velocity

Velocity is the derivative of position:

$$\begin{aligned}v(t) &= \frac{d}{dt} [t^3 - 3t^2 + 2t + 5] \\&= 3t^2 - 6t + 2 \\v(2) &= 3 \times (2^2) - 6 \times (2) + 2 \\&= 2 \text{ m/s.}\end{aligned}$$

Acceleration

Acceleration is the derivative of velocity:

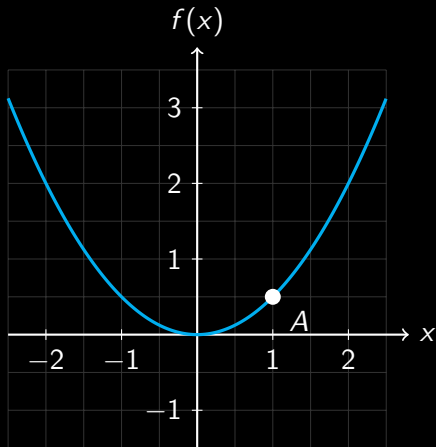
$$a(t) = \frac{d}{dt} [3t^2 - 6t + 2]$$

$$= 6t - 6,$$

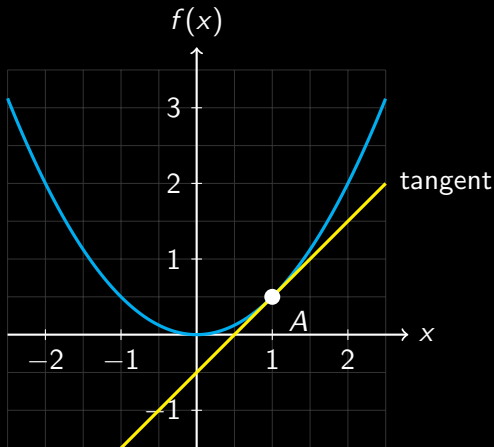
$$a(2) = 6 \times (2) - 6$$

$$= 6 \text{ m/s}^2.$$

Tangent and Normal Lines

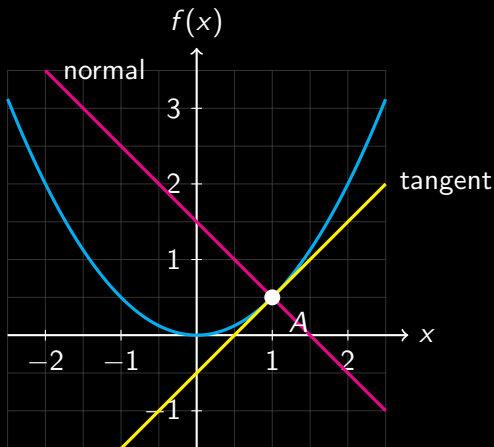


Tangent and Normal Lines



The tangent passes through A and has the same gradient as the function at A .

Tangent and Normal Lines



The normal passes through A and is perpendicular to the tangent.

Finding the Tangent Line

Find the tangent to $y = x^3$ at $x = 1$.

$$y = 1^3 = 1, \quad A = (1, 1),$$

$$\frac{dy}{dx} = 3x^2,$$

$$m_{\text{tangent}} = \left. \frac{dy}{dx} \right|_{x=1} = 3.$$

Using $y - y_1 = m(x - x_1)$,

$$y - 1 = 3(x - 1)$$

$$y = 3x - 2.$$

Finding the Normal Line

Find the normal to $y = x^3$ at $x = 1$.

The normal is perpendicular to the tangent, so

$$\begin{aligned} m_{\text{normal}} &= -\frac{1}{m_{\text{tangent}}} \\ &= -\frac{1}{3}. \end{aligned}$$

Using $A = (1, 1)$ and the point-slope form,

$$\begin{aligned} y - 1 &= -\frac{1}{3}(x - 1) \\ y &= -\frac{1}{3}x + \frac{4}{3}. \end{aligned}$$

x -intercept of the Tangent Line

The tangent is $y = 3x - 2$.

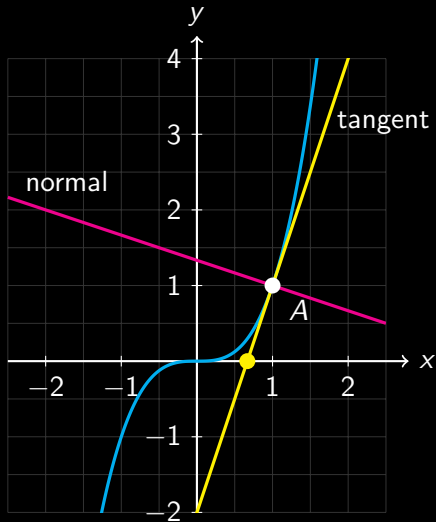
At the x -intercept, $y = 0$.

$$0 = 3x - 2$$

$$x = \frac{2}{3}.$$

The tangent has an x -intercept at $x = \frac{2}{3}$.

Function, Tangent, and Normal



The tangent and normal meet at A and are perpendicular.

Finding the Tangent Line

Find the tangent to $y = \sin(x)$ at $x = \frac{\pi}{3}$.

$$y = \sin\left(\frac{\pi}{3}\right) = \frac{\sqrt{3}}{2}, \quad A = \left(\frac{\pi}{3}, \frac{\sqrt{3}}{2}\right),$$

$$\frac{dy}{dx} = \cos(x),$$

$$m_{\text{tangent}} = \left. \frac{dy}{dx} \right|_{x=\pi/3} = \cos\left(\frac{\pi}{3}\right) = \frac{1}{2}.$$

$$\begin{aligned} y - \frac{\sqrt{3}}{2} &= \frac{1}{2} \left(x - \frac{\pi}{3}\right) \\ &\approx \frac{x}{2} + 0.34 \end{aligned}$$

Finding the Normal Line

Find the normal to $y = \sin(x)$ at $x = \frac{\pi}{3}$.

The normal is perpendicular to the tangent, so

$$\begin{aligned} m_{\text{normal}} &= -\frac{1}{m_{\text{tangent}}} \\ &= -2. \end{aligned}$$

Using $A = \left(\frac{\pi}{3}, \frac{\sqrt{3}}{2}\right)$,

$$\begin{aligned} y - \frac{\sqrt{3}}{2} &= -2 \left(x - \frac{\pi}{3}\right) \\ &\approx -2x + 2.96 \end{aligned}$$

x -intercept of the Tangent Line

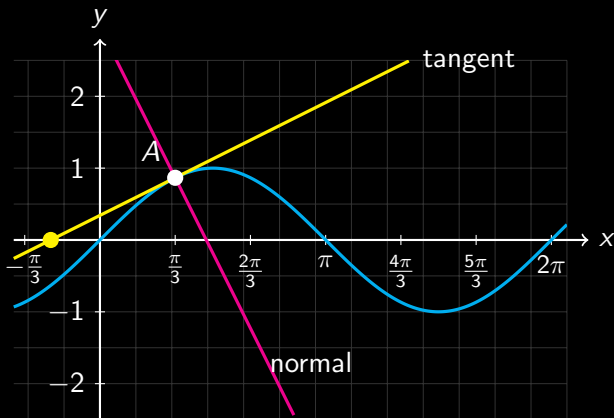
At the x -intercept of the tangent, $y = 0$:

$$\frac{x}{2} + 0.34 = 0$$

$$x = -0.68.$$

The tangent has an x -intercept at $x = -0.68$.

Function, Tangent, and Normal



$y = \sin(x)$, its tangent, and its normal at $x = \frac{\pi}{3}$.